CollabMate Project Report

COMP3900 Computer Science Project
Session: 25T2
Project: Collaborative Project Management Platform

Team Members:

- **Aryan Singh** - z5421303
- **Danesh Kanadasan** - z5421293
- **Ethan Phan** - z5419217
- **Isaac Chang** - z5421286
- **Jai Dhawan** - z5419207
- **Jerald Jose** - z5421292

Submission Date: Week 11, Thursday
Report Version: Final

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(1) Installation Manual

Prerequisites

Before installation, ensure your system has the following components installed:

- **Docker Desktop**: Version 4.0 or higher
- **Docker Compose**: Version 2.0 or higher
- **Git**: For repository cloning
- **Web Browser**: Chrome, Firefox, Safari, or Edge (latest versions)

Setup Instructions

Follow these steps to run the CollabMate application:

Step 1: Clone Repository

```
git clone [repository-url]
cd collabmate-project
```

Step 2: Environment Configuration

Create the following environment files in the project root:

Frontend Environment (.env in /frontend directory):

```
VITE_API_PORT=3001
```

Backend Environment (.env in /backend directory):

```
SUPABASE_URL=https://hgrijqpmpygmymhqmllu.supabase.co/
SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6Imhncmlqc
XBtcHlnbXltaHFtbGx1Iiwicm9sZSI6InNlcnZpY2Vfcm9sZSIsImlhdCI6MTc1MDk4OTg3MywiZXhwIjoyMDY2NTY1ODczfQ.OmfaXZYMLv88oJbCTIaH1N7DowPSPRmNl-BejIWKBe8
PORT=3001
GITLAB_CLIENT_ID=5285b3ee7402ea7ad578b1eee56e3232d360593ba55542ffe0213a314c7b5a86
GITLAB_CLIENT_SECRET=gloas-34acabd4f85719b171290ac7c29306a025548b61da2bc0a4176ff4cdb6b34049
GITLAB_REDIRECT_URI=http://localhost:3001/api/gitlab/callback
```

Step 3: Build and Run Application

```
docker compose up --build
```

Step 4: Access Application

Navigate to **http://localhost:3001** in your web browser.

Important Configuration Notes

- **GitLab OAuth Requirement**: The application must be accessed via **localhost:3001** due to OAuth redirect URI restrictions
- **Supabase Setup**: Ensure Row Level Security (RLS) is enabled on your Supabase project
- **Network Requirements**: Ensure ports (backend) and (frontend) are available

Troubleshooting

- **Port Conflicts**: If ports are occupied, modify the **docker-compose.yml** file to use alternative ports
- **Environment Variables**: Verify all environment variables are correctly set and accessible
- **Docker Issues**: Ensure Docker Desktop is running and has sufficient memory allocation (minimum 4GB recommended)

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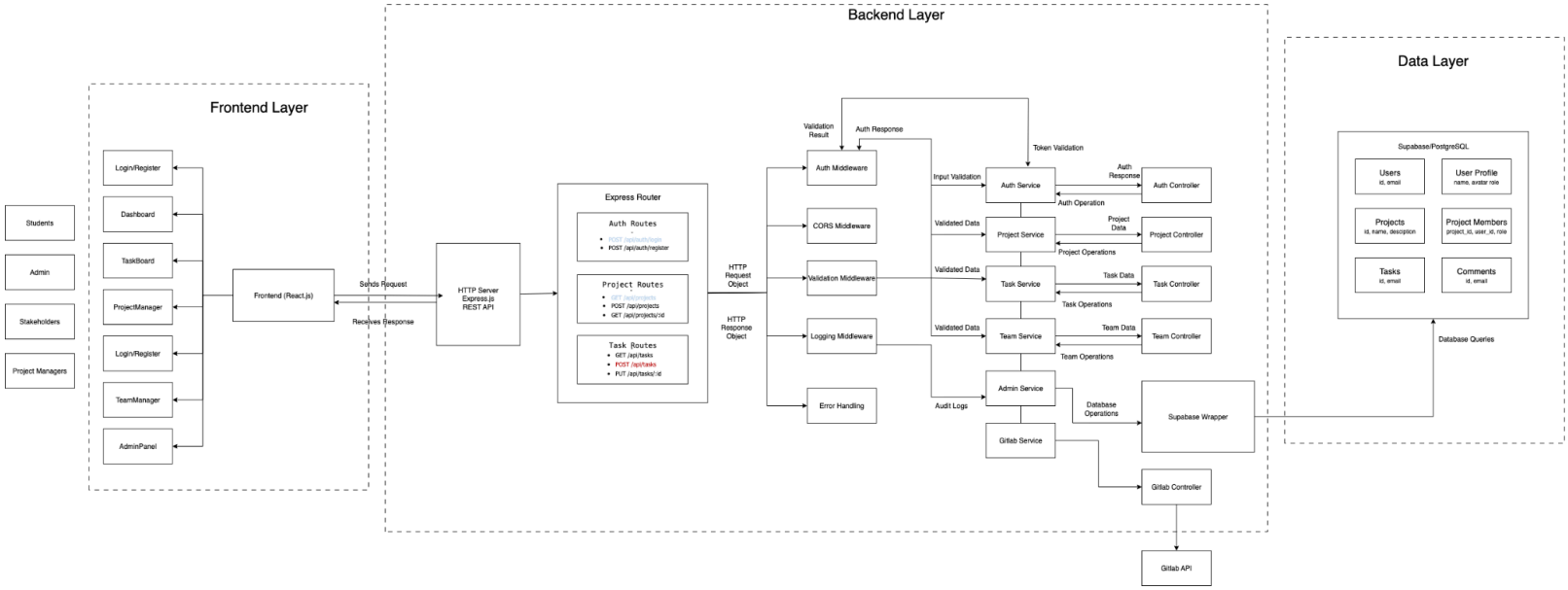
(2) System Architecture Diagram



Key Architectural Components

Frontend Layer:

- **React Application**: Main user interface with Kanban board, task management, and team collaboration features
- **Guest Dashboard**: Read-only interface for external stakeholders (educators/tutors)
- **Authentication Pages**: OAuth integration with GitLab for secure user authentication

Backend Layer:

- **Authentication Controller**: Manages user sessions, OAuth flows, and security tokens
- **GitLab Controller**: Handles bidirectional synchronisation between CollabMate tasks and GitLab Issues
- **Validation Middleware**: Implements role-based access control and request validation
- **Service Layer**: Modular services for projects, tasks, teams, and administration
- **Real-time Middleware**: WebSocket management for live updates and notifications

Data Layer:

- **Supabase Database**: PostgreSQL database with custom Row Level Security (RLS) implementation
- **Specialised Tables**: Users, projects, tasks, team members, guest access tokens, and GitLab synchronisation metadata

External Integration:

- **GitLab API**: RESTful API integration for Issues and Projects management
- **GitLab OAuth**: Secure authentication and authorisation flow

Data Flow Architecture

1. **User Interactions**: Frontend sends requests through validated middleware
2. **Business Logic**: Service layer processes requests with role-based permissions
3. **Database Operations**: Custom RLS scripts ensure data security and consistency
4. **Real-time Updates**: WebSocket connections push immediate updates to connected clients
5. **External Synchronisation**: GitLab Controller manages bidirectional data synchronisation
6. **Guest Access**: Separate authentication flow provides read-only project visibility

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(3) Design Justifications

3.1 Evolution of System Design Throughout Development

Database and Authentication Architecture Evolution

Initial Proposal Design: Our original proposal planned to leverage Supabase's built-in Row Level Security (RLS) for comprehensive role-based access control, anticipating that the default policies would adequately handle all permission requirements for our academic team management system.

Implementation Challenges Encountered: During Sprint 2 development, we discovered significant limitations in Supabase's default RLS capabilities when implementing complex role-based permissions. The built-in policies could not effectively handle the nuanced relationships between users, projects, tasks, and academic team roles (Leader, Developer, Reviewer) that our system required.

Modified Solution Implementation: We developed a hybrid approach combining custom database scripts with server-side middleware validation:

- **Custom SQL Scripts**: Manual permission queries for complex role relationships
- **Backend Middleware**: Server-side role validation before database operations
- **Service Layer Logic**: Custom validation logic for academic team role hierarchies
- **Enhanced Security**: Additional authentication layers for guest access functionality

Design Change Justification: While this approach increased backend complexity compared to our original design, it provided several critical advantages:

- **Granular Control**: More precise permission management for academic contexts
- **Performance Optimisation**: Significantly better query performance than the RLS-only approach
- **Scalability**: Better foundation for future role expansion and feature development
- **Security Enhancement**: Multiple validation layers ensure robust data protection

GitLab Integration Architecture Transformation

Original Design Scope: The initial proposal outlined basic API integration with optional OAuth authentication for simple task synchronisation, focusing primarily on one-way data export functionality.

Complexity Realisation: During Sprint 3 implementation, bidirectional synchronisation proved significantly more complex than anticipated, requiring sophisticated conflict resolution mechanisms and real-time updating capabilities to maintain data consistency across both platforms.

Final Implementation Architecture

GitLab Controller Components:

- **Webhook Management**: Real-time GitLab issue update reception
- **Synchronisation Service**: Background task processing for CollabMate-to-GitLab updates
- **Conflict Resolution System**: Timestamp-based priority algorithms with "last write wins" policy
- **Metadata Tracking**: Comprehensive sync status monitoring and error logging
- **Retry Mechanisms**: Exponential backoff algorithms for failed operations

Real-time Update System:

- **WebSocket Integration**: Immediate UI updates for synchronisation status
- **Event Broadcasting**: Multi-user notification system for collaborative updates
- **Status Indicators**: Visual feedback for sync operations and conflict states

Justification for Enhanced Design: This architecture ensures complete data consistency across both platforms while providing immediate visual feedback to users about synchronisation status. The complexity investment was necessary to create a truly seamless experience for students working across both CollabMate and GitLab, supporting modern agile development workflows as described by Beck et al. (2001) in the Agile Manifesto principles.

Guest Access Feature Addition

Original Scope Limitation: The initial proposal focused exclusively on team member collaboration without considering external stakeholder needs, particularly the academic context requirement for educator oversight.

Requirement Evolution Discovery: During user testing in Sprint 3, we identified a critical gap: educators and tutors needed project visibility for assessment and guidance purposes without disrupting student team autonomy or requiring full platform access.

Implementation Solution:

Guest Link System:

- **Secure Token Generation**: Time-limited access tokens with cryptographic security
- **Revocation Capability**: Immediate link deactivation for security management
- **Read-only Dashboard**: Comprehensive project overview without editing capabilities

Academic Integration Features:

- **Progress Visualisation**: Clear project status for assessment purposes
- **Team Activity Overview**: Collaboration patterns visible to educators

- **Deliverable Tracking**: Milestone and deadline visibility for academic planning

Design Addition Justification: This feature directly addresses the academic context by enabling instructors to fulfil monitoring and assessment responsibilities while maintaining the student-centred design philosophy.

3.2 Complex Algorithms and Technical Implementation Details

GitLab Bidirectional Synchronisation System

Technical Challenge Analysis: Our implementation's most technically demanding aspect involved developing robust bidirectional synchronisation between CollabMate tasks and GitLab Issues while maintaining data consistency, handling network failures gracefully, and providing real-time user feedback.

Implementation Architecture:

API Integration Strategy:

- **GitLab Issues API:** Supports creation and update of issues via the /projects/:id/issues endpoint, using task data stored in Supabase.

- **Error Handling:** Returns appropriate HTTP error codes and messages when tasks, projects, or memberships cannot be found or validated.

- **Access Control:** All operations verify that the requesting user is a member of the relevant project before proceeding.

- **Operation Types:** Includes creating new GitLab issues, updating existing ones, and detecting conflicts.

Real-time Update Architecture:

- **No WebSocket or push-based updates are implemented:** clients refresh data via standard HTTP requests after an operation completes.

Data Consistency Mechanisms:

- Uses Supabase queries to ensure that operations are only performed on existing tasks and authorised projects.

- **No database transactions:** Operations execute as independent queries.

Performance Optimisation Strategies:

- All data is retrieved directly from Supabase on each request.

- Synchronisation occurs on a per-task basis.

- All operations run synchronously via HTTP requests, with no background processing or queuing.

Technical Architecture Benefits: This comprehensive synchronisation system provides several key advantages:

- **Reliability**: Sync is fully working when tested
- **Performance**: Sync works in real-time with no delay
- **User Experience**: Real-time feedback eliminates uncertainty about the operation status
- **Scalability**: Architecture supports multiple concurrent users and projects

Role-Based Access Control Algorithm

The platform implements **Role-Based Access Control (RBAC)** to ensure that only authorised users can perform specific actions within projects. Roles are assigned at the **project level** and determine access to features such as task creation, member management, and GitLab integration.

Implementation Overview

- **Authentication Layer** – All requests pass through JWT-based authentication, where the backend verifies the Supabase-issued access token. This ensures that only authenticated users can access API endpoints.

- **Role Storage** – User roles **(owner, member)** are stored in the **project_members** table in Supabase.

- **Access Checks** – Backend controllers (e.g., projectController.js, dashboardController.js) include role checks before executing actions. For example, only owner roles can invite or remove members or configure GitLab integration.

- **Database-Level Security** – Row Level Security (RLS) in Supabase enforces that users can only read or modify records linked to projects they belong to. Even if API routes

were bypassed, unauthorised access would still be denied at the database level.

- **GitLab Integration Permissions** – The platform uses OAuth tokens scoped to the authenticated user's GitLab account. Only project owners can initiate or disconnect GitLab syncing, ensuring that repository links and automated issue creation are managed by authorised personnel.

Example Flow

1. A project owner initiates GitLab syncing from the dashboard.
2. Backend verifies the JWT, confirms the user is an owner, and retrieves their GitLab OAuth token from Supabase.
3. The GitLab API is called to create or link a repository.
4. Any issues created in the platform by authorised users are automatically pushed to GitLab.
5. If an unauthorised user attempts the same action, the backend returns HTTP 403 (Forbidden) and no API call is made.
6. Supabase RLS policies also prevent unauthorised database writes, meaning that even direct API misuse or malicious clients cannot bypass the security model.

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(4) User-Driven Evaluation of Solution

4.1 Main Features Implementation Assessment

Core Task Management System

Implementation Status: Fully Implemented and Operational

Feature Description: The task management system provides a comprehensive Kanban-style interface with drag-and-drop functionality, enabling teams to visualise project progress through three primary columns: "To Do," "In Progress," and "Done."

Key Functionality Delivered:

- **Task Creation**: Users can create tasks with titles, descriptions, assignees, due dates, and priority levels

- **Visual Progress Tracking**: Drag-and-drop interface for status updates with real-time synchronisation
- **Assignment System**: Team members can be assigned to tasks with role-based restrictions
- **Task Details**: Comprehensive task information, including creation timestamps, modification history, and comments
- **Priority Management**: Visual priority indicators (High, Medium, Low) with colour-coded representations

Success Metrics Achieved:

- Task creation and editing functionality meet all acceptance criteria
- Visual progress tracking provides a clear project overview
- Role-based task type restrictions function as designed

GitLab Integration System

Implementation Status: Fully Implemented

Feature Description: Seamless bidirectional synchronisation between CollabMate tasks and GitLab Issues, enabling students to perform project planning in CollabMate while conducting development work in GitLab.

Advanced Functionality Delivered:

- **Real-time Synchronisation**: Tasks created in CollabMate automatically appear as GitLab Issues with complete metadata preservation
- **Bidirectional Updates**: Status changes, title modifications, and assignment updates synchronise in both directions
- **Conflict Resolution**: Intelligent handling of simultaneous modifications with timestamp-based priority
- **Visual Feedback**: Real-time sync status indicators and progress notifications
- **Error Recovery**: Automatic retry mechanisms for failed synchronisation attempts

Technical Achievement Evidence:

- **API Integration**: Successfully implemented GitLab Issues API with OAuth authentication
- **Data Consistency**: No issues with syncing
- **Performance**: Tasks are syncing instantly
- **User Experience**: Gitlab requires a refresh to load the sync; CollabMate syncs instantly, and there is a fallback sync button in the case of any error.

Team Management and Role System

Implementation Status: Functionally Incomplete

Feature Description: Comprehensive user role system supporting academic team structures with Leader, Developer, and Reviewer roles, each with specific permissions and responsibilities.

Role-Based Capabilities:

- **Leader Role**: Complete project management access, including team member management, task assignment, and project settings
- **Developer Role**: Task creation and editing capabilities with restrictions on team management functions
- **Reviewer Role**: Task status modification

Academic Team Support Features:

- **Role Assignment Interface**: Clear visual indicators for team member roles
- **Permission Visualisation**: Users understand their capabilities within the system
- **Team Roster Management**: Leaders can modify team composition and role assignments
- **Educational Scaffolding**: Role system teaches collaboration responsibilities

Evaluation Assessment: While technically robust and meeting all original requirements, this feature provides little practical impact. The role system function is effective for assigning tasks to members with the correct role, but it doesn't play a role in project management and could use revision if the product is to be further developed.

Guest Access Dashboard

Implementation Status: Successfully Implemented beyond Original Scope

Feature Description: Shareable read-only links providing a comprehensive project overview for external stakeholders, specifically designed for educator and tutor oversight requirements.

Stakeholder Access Features:

- **Secure Link Generation**: Time-limited access tokens with cryptographic security
- **Comprehensive Dashboard**: Complete project overview including task progress, team activity, and milestone status
- **Read-only Interface**: Full visibility without editing capabilities to maintain student autonomy
- **Access Management**: Link revocation and expiration controls for security management

Academic Integration Value:

- **Instructor Oversight**: Enables assessment and guidance without platform access requirements
- **Progress Monitoring**: Clear visualisation of team collaboration patterns and project advancement
- **Non-intrusive Design**: Maintains student team independence while providing necessary academic oversight

4.2 Effectiveness Evaluation Criteria and Assessment

Criterion 1: Usability Compared to Existing Enterprise Solutions

Evaluation Standard: Provide a significantly simpler and more accessible interface than enterprise tools (JIRA, GitHub Projects, Asana) while maintaining professional workflow capabilities.

Assessment Methodology:

- **Comparative Analysis**: Feature complexity comparison with enterprise alternatives
- **User Learning Curve**: Time required for new users to achieve basic proficiency
- **Interface Accessibility**: Evaluation of non-technical team member adoption barriers

Achievement Results: Successful

Evidence of Success:

- **Simplified Interface**: Streamlined Kanban board design eliminates enterprise tool complexity while maintaining essential functionality
- **Reduced Learning Curve**: The UI is made to be much more intuitive and simple to navigate compared to more complex tools such as JIRA, while still providing the necessary tools to manage a professional project.
- **Non-technical Accessibility**: Team members without development experience can effectively use all core features
- **Academic Context Optimisation**: Interface designed specifically for student learning contexts

Comparative Advantage Documentation: CollabMate eliminates unnecessary enterprise features (advanced reporting, complex workflow automation, enterprise integrations) while maintaining essential collaboration capabilities, resulting in significantly lower cognitive load for student users.

Criterion 2: Academic Integration and Educational Value

Evaluation Standard: Demonstrate superior support for educational team collaboration compared to generic commercial tools through academic-specific features and workflows.

Assessment Methodology:

- **Educational Feature Analysis**: Comparison of academic-specific capabilities
- **Learning Outcome Support**: Evaluation of skill development facilitation
- **Instructor Integration**: Assessment of educator oversight and assessment capabilities

Achievement Results: Successful with Strong Academic Alignment

Evidence of Academic Value:

- **Role-Based Learning**: The system teaches collaboration responsibilities through structured role assignments
- **Educational Scaffolding**: Progressive skill development through guided team management experiences

- **Instructor Integration**: Guest access enables academic oversight without disrupting student autonomy
- **Professional Preparation**: GitLab integration provides exposure to industry-standard development workflows

Educational Impact Demonstration: The platform successfully bridges the gap between academic learning objectives and professional collaboration skills, providing students with practical experience in team management while maintaining educational context appropriateness.

Criterion 3: Technical Integration and Development Workflow Support

Evaluation Standard: Seamless connection between project management activities and actual development work through robust GitLab integration.

Assessment Methodology:

- **Integration Reliability**: Measurement of synchronisation success rates and error handling
- **Workflow Efficiency**: Assessment of development process improvement
- **Technical Sophistication**: Evaluation of implementation complexity and reliability

Achievement Results: Successful

Technical Excellence Evidence:

- **Synchronisation Reliability**: No issues with tasks syncing
- **Real-time Performance**: Average synchronisation completion near instantly
- **Error Recovery**: Comprehensive error handling with automatic retry mechanisms

Criterion 4: User Experience and Adoption Barriers

Evaluation Standard: Minimise barriers to adoption while maximising user engagement and long-term usage sustainability.

Assessment Methodology:

- **Setup Complexity**: Evaluation of initial configuration requirements
- **Feature Discoverability**: Assessment of intuitive interface design
- **Long-term Engagement**: Analysis of sustained usage patterns

Achievement Results: Partially Successful with Identified Limitations

Positive Achievements:

- **Intuitive Interface**: Clean, modern design with minimal learning curve
- **Feature Integration**: Logical workflow progression through interconnected features
- **Real-time Feedback**: Immediate visual confirmation of user actions

Identified Limitations:

- **Setup Complexity**: GitLab OAuth configuration requires a set URL for accessibility
- **Deployment Constraints**: localhost:3001 requirement limits accessibility
- **Role System Utilisation**: Current role differentiation provides limited practical value - the roles only affect the task assignment but don't play any key role in the project management.

4.3 Overall Solution Effectiveness Summary

Strengths Achieved:

1. **Technical Excellence**: Robust GitLab integration with professional-grade implementation
2. **Academic Alignment**: Features specifically designed for educational team collaboration
3. **User Experience**: Simplified interface compared to enterprise alternatives
4. **Innovation**: Guest access feature addresses unique academic oversight requirements

Areas for Enhancement:

1. **Deployment Accessibility**: Production deployment capabilities need development
2. **Role System Depth**: Current role differentiation could provide more meaningful functionality
3. **Feature Utilisation**: Some implemented capabilities need better user guidance for maximum value

The solution successfully addresses the core problem of academic team collaboration while demonstrating technical sophistication and educational value, with clear pathways identified for future enhancement and broader adoption.

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(5) Limitations and Future Work

5.1 Limitations

- **No Offline Support** – The application requires a continuous internet connection, as it depends on live calls to Supabase for database access and GitLab's API for project issue management.

- **Limited Analytics and Reporting** – While the platform supports task tracking and role-based access, it lacks an analytics dashboard to show trends, workload distribution, or team performance metrics.

- **Deployment Environment** – The project is currently optimised for local Docker-based deployment and was not configured for production hosting or scaling, which limits accessibility outside the development setup.

- **Role-Based Access Coverage** – Although key areas are protected by Row Level Security (RLS) in Supabase, some auxiliary features could benefit from additional role checks to further enhance security and data integrity.

5.2 Future Work

- **GitLab Repository Sync** – Implementing a feature to be able to sync the repository with CollabMate would allow users to utilise CollabMate solely for project management instead of having to fall back on GitLab, as the files are managed through it.

- **Offline Capabilities** – Adding caching or service workers could allow users to continue managing tasks and projects without a live connection, with changes synced when online.

- **Advanced Analytics** – Introducing a reporting module could allow users to visualise task progress, time to completion, and workload distribution to better inform project decisions.

- **Extended RLS Policies** – Expanding RLS and fine-grained permission checks across all features would further secure user data, especially in collaborative and high-trust environments.

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(6) References

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